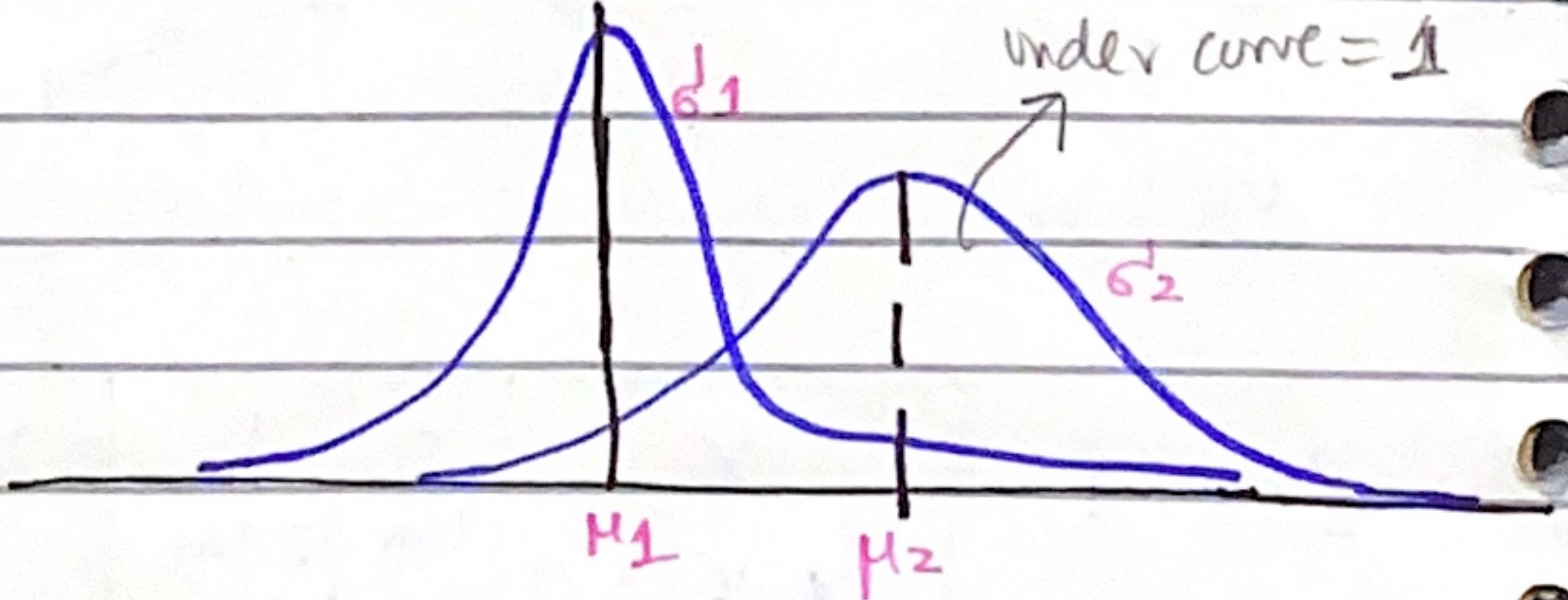
LECTURE - 19

Friday
10/31/26.

Normal Distribution: For Continuous random variable. Total Area under curve = 1

Prob Density Func:

$$\underbrace{n(x;\mu,\sigma^2)}_{\text{parameters}} = \frac{e^{-\frac{1}{2\sigma^2}\cdot(x-\mu)^2}}{\sqrt{2\pi\sigma}}$$



- μ controls horizontal location on x-axis
- σ controls expansion/compression of hill.

$\mu \to$ mean

$\sigma \to$ SD.

$\mu_1 < \mu_2$ & $\sigma_1 < \sigma_2$

- Putting the value of Z in Φ func. gives us probability where Φ is a func. mapping Z onto graph whose area under curve gives prob.

'Z'

FORMULAS:

① $P(X \le a) = \Phi\left(\frac{a-\mu}{\sigma}\right)$

② $P(X \ge a) = 1 - \Phi\left(\frac{a-\mu}{\sigma}\right)$

③ $P(a \le X \le b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$

⇒ Finding Probability from given X, μ, σ:

Q: Scores of Students are normally distributed with mean $\mu = 500$, SD $\sigma = 100$. What percentage of candidates recieve scores:

(a) less than 400

converting X into Z

$P(X < 400) \rightarrow$ Case ① $= P\left[\frac{X-\mu}{\sigma} < \frac{400-500}{100}\right]$

$= P(Z < -1) = \Phi(-1)$

★ To commute $\Phi(-1)$; go in Z-score table & find value which has Z-score exactly or a bit higher than -1.00.

$\Phi(-1) = 0.15866 \longrightarrow$ answer $= \boxed{15.87\%}$

(b) More than 700.

$P(X > 700) = 1 - \Phi\left(\frac{700-500}{100}\right)$

from table

$= 1 - \Phi(2) \longrightarrow 1 - 0.9772$

answer $= 0.02275 = \boxed{2.28\%}$

(c) which differs by more than 150 from mean (μ).

$P(|X-\mu| > 150) = 1 - P(|X-\mu| < 150)$

modulus since 150 above & below $= 1 - P(-150 < X-\mu < +150)$ removing modulus.

$= 1 - P(-150+\mu < X < 150+\mu)$

$= 1 - P(350 < X < 650)$

$P(350 < X < 650) = \Phi[(650-500)/100] - \Phi[(350-500)/100]$

$= 0.9332 - 0.0662$

$= 0.8664$

$1 - 0.8664 = \boxed{13.36\%}$

⇒ Finding X for given probability:

Q: If $X \sim N(70, 25)$ find
(Normally distributed → N, μ → 70, σ^2 → 25)

(a) a point that has 87.9% of distribution below it.

$P(X < a) = 0.879$ $\qquad \mu = 70$

$P\left(Z < \frac{a-70}{5}\right) = 0.879$ $\qquad \sigma = \sqrt{25} = 5$

$\Phi\left(\frac{a-70}{5}\right) = 0.879$

$\Phi^{-1}(0.879) = (a-70)/5$

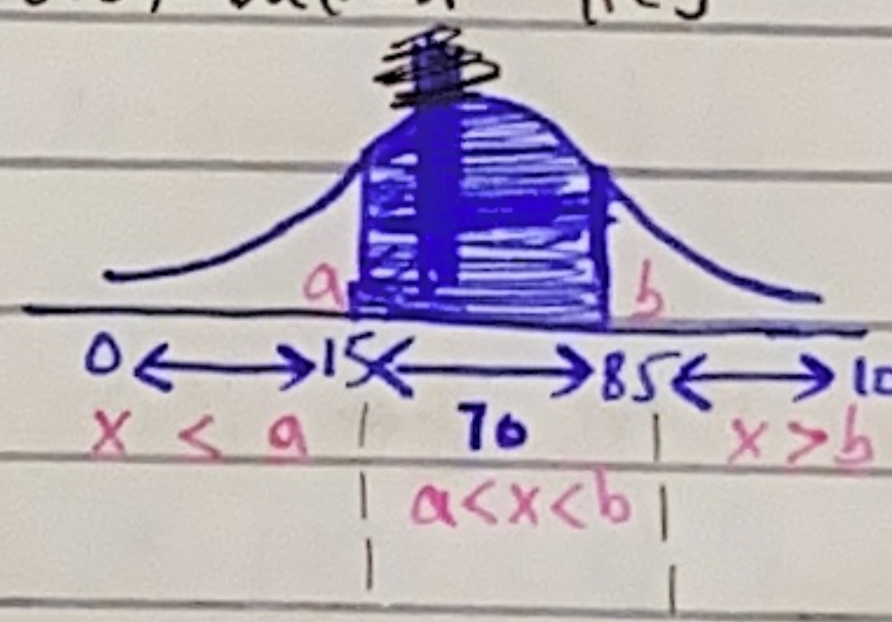
* reverse track from table $= 1.17$

$1.17 = (a-70)/5$

$\boxed{a = 75.85}$

(b) Two such points btw which the central 70% of distribution lies.

Since central 70% ∴ 15% percent from each side will be left. ∴



$P(X<a) + P(a<X<b) + P(X>b) = 1$

$P(X<a) + P(X>b) = 1 - P(a<X<b)$

$= 1 - 0.70$

$= 0.30$

* since $P(X<a) = P(X>b)$ as symmetric ∴ each is of 0.15 (0.3/2)

$P(X<a) = 0.15$

$\Phi\left(\frac{a-70}{5}\right) = 0.15$

$a = (\Phi^{-1}(0.15) * 5) + 70$

$a = \boxed{64.818}$

$P(X>b) = 0.15$

$P(X<b) = 1 - 0.15 = \boxed{0.85}$

$\Phi^{-1}(0.85) = \frac{b-70}{5}$

$b = \boxed{75.182}$

⟹ If Ques. ask 33rd percentile / Decile-2 / Q3 :

Then we'll just do same as before.

$P(X < a) = 0.33$ OR 0.2 OR 0.75

33p D2 Q3

⟹ Finding μ & σ for given X & prob.

Q. In normal distri. 33% of values are under 48 & 12.3% are over 60. Find mean & SD.

(P_1 = 33%, X_1 = 48, P_2 = 12.3%, X_2 = 60)

$P(X < 48) = 0.33$

$\phi^{-1}(0.33) = \frac{48 - \mu}{\sigma} \rightarrow (A)$

$P(X > 60) = 0.123$

$P(X \cancel{>} 60) = 1 - 0.123$

$= 0.877$

$\phi^{-1}(0.877) = \frac{60 - \mu}{\sigma} \rightarrow (B)$

★ Solve (A) & (B) simultanously to get μ & σ.

- Mean(μ) = mode = Median as symmetric graph
- variance = σ^2

Standard Normal Distribution has $\mu = 0$, $\sigma = 1$.

- $q_1 = \mu - (0.6745 * \sigma)$
- $q_3 = \mu + (0.6745 * \sigma)$

☐ Normal Approximation to Binomial distri:

(continuous) (discrete)

- If we've $N = 100$ with $p = 0.5$ (success in Binomial) & need to find $P(x < 27)$ then we do; $P(x<27) = P(x=0) + P(x=1) \cdots P(x=26)$.
  This is a really large calculation.
- Therefore we'll see if $np > 5$ which in this case is as $100 * 0.5 = 50$ & $50 > 5$; then we'll use NORMAL DISTRIBUTION.

where $X \sim N(\overset{\mu}{np}, \overset{\sigma^2 = \text{variance}}{npq})$ and then:

$$P(X<27) = \phi\left(\frac{27 - \overset{np}{\mu}}{\underset{\sqrt{npq}}{\sigma}}\right)$$

⇒ Since we're converting Discrete (Binomial) into Continuous (Normal);
∴ we need a correction value for accurate results:

Continuity Correction:

	Discrete	Continuous
1	$P(X=a)$	$P(a-0.5 < X < a+0.5)$
2	$P(X<=a)$	$P(X <= a+0.5)$
3	$P(X<a)$	$P(X < a-0.5)$
4	$P(X>a)$	$P(X > a+0.5)$
5	$P(X \geq a)$	$P(X \geq a-0.5)$

Q: Coin tossed 20 times; find prob. that no. of Heads is b/w 10 & 14 inclusive by using suitable method.

Ans: $n = 20$, $p = 0.5$ (heads) → $20 * 0.5 = n \cdot p$
$10 = np$

* since $np > 5$ ∴ we'll use Normal Distribution.

• At $X = 10$ & 14 we'll do $10-0.5$ & $14+0.5$ (correction)
$= 9.5$ & 14.5 (rule 1 of Continuity)

$X = 9.5 \rightarrow Z = \frac{9.5-\mu}{\sigma}$, $X = 14.5 \rightarrow Z = \frac{14.5-\mu}{\sigma}$

$\mu = np = 10$, $\sigma = \sqrt{npq} = 2.24$

$X = 9.5 \rightarrow \boxed{Z = -0.22}$ $X = 14.5 \rightarrow \boxed{Z = +2.01}$

Now; Normal Distri.

$$P(10 \leq X \leq 14) = P(\underset{a}{-0.22} \leq Z \leq \underset{b}{2.01})$$
$$= \phi(2.01) - \phi(-0.22)$$
$$= 0.9778 - 0.4129$$

ans $= \boxed{0.5649}$